May/June 2017 Solutions

Question 8 (Kinematics):
Given displacement $s = t^3 - (52/2)t^2 - 2t$.
(i) Velocity $v = ds/dt = 3t^2 - 52t - 2$.
At $t = 3.5$s, $v = 3(3.5)^2 - 52(3.5) - 2 = 36.75 - 182 - 2 = -147.25$ m/s.

(ii) Momentarily at rest means $v = 0 \rightarrow 3t^2 - 52t - 2 = 0$.
Using the quadratic formula: $t = [52 \pm \sqrt{(-52)^2 - 4(3)(-2)}] / (2*3)$
$t = [52 \pm \sqrt{2704 + 24}] / 6 = [52 \pm 52.23] / 6$
Since $t \geq 0$, $t = 104.23 / 6 = 17.37$ seconds.

May/June 2023 Solutions

Question 1 (Algebra & Functions):
(a) Solve $3^{(2x+1)} - 5(3^x) - 2 = 0$.
Let $u = 3^x$, then $3(u^2) - 5u - 2 = 0$.
Factorizing: $(3u + 1)(u - 2) = 0 \rightarrow u = -1/3$ or $u = 2$.
Since 3^x cannot be negative, $3^x = 2$.
Taking logs: $x = \log(2) / \log(3)$ approx 0.631.

(b) Given $3x+2$ is a factor of $3x^3 + bx^2 - 3x - 2$:
By Remainder Theorem, $f(-2/3) = 0$.
$3(-2/3)^3 + b(-2/3)^2 - 3(-2/3) - 2 = 0$
$-8/9 + 4b/9 + 2 - 2 = 0 \rightarrow 4b/9 = 8/9 \rightarrow b = 2$.

May/June 2024 Solutions

Question 1 (Polynomials & Quadratics):
(a) (i) Given $x - 2$ is a factor of $3x^3 + 8x^2 - 20x - 16$.
By long division or synthetic division, the quotient is $3x^2 + 14x + 8$.
Factorizing the quadratic: $(3x + 2)(x + 4)$.
The other linear factors are $(3x + 2)$ and $(x + 4)$.

(ii) Simplify: $[(3x^3 + 8x^2 - 20x - 16) / (x^2 - 4)] * [(x + 2) / (x + 4)]$
Substitute the factors: $[(x - 2)(3x + 2)(x + 4) / ((x - 2)(x + 2))] * [(x + 2) / (x + 4)]$
Cancelling matching terms from top and bottom gives the final simplified result: $3x + 2$.